

## Universal Worksheet Engine Showcase

A pdfLaTeX-safe portfolio of printable classroom artefacts

This portfolio demonstrates one engine producing structured worksheets across primary learning, STEM, HSC-style mathematics, science practicals, humanities analysis, computing, diagrams, graphs, circuits, chemical notation, answer keys, and QR-supported teacher workflows.

**Teacher draft  
review before use**

Design principle: notation, diagrams, worked space, and marking guidance are not decoration; they are part of the pedagogy.

### Capability Map

- Primary visual learning
- Measurement and ruler work
- Geometry and transformations
- Statistics, probability, networks
- HSC calculus and vectors
- Science practical design
- Physics, chemistry, biology
- Humanities, English, business
- Computing, logic and algorithms
- Teacher answer key and QR support

### Safe Engineering Choices

- One pdfLaTeX document
- No external images
- No fontspec
- Sans-serif via `\sfdefault`
- TikZ/PGF graphics only
- White-backed labels
- Bounded graphs and diagrams
- Footer reinforces review

### Pedagogical Coverage

- Access starters
- Representation shifts
- Misconception checks
- Parameter shifts
- Practical investigation skills
- Exam-style prompts
- Proof and generalisation
- Multi-step synthesis

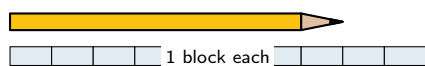
Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Part A: Universal Cross-Curricular Sampler

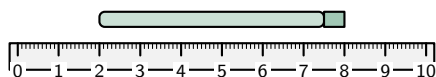
Dense layouts, diagrams, graphs, symbolic notation, practical skills and answer spaces in one coherent printable system.

1. **Primary Measurement:** Count the blocks. How long is the pencil? [1 mark]

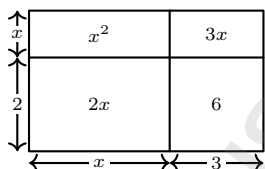


The pencil is \_\_\_\_\_ blocks long.

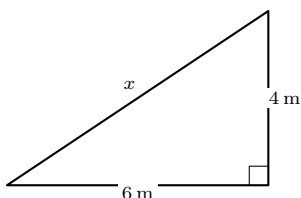
2. **Spot the Measuring Error:** A student says the marker is 5 cm long. Explain the error and give the correct length. [2 marks]



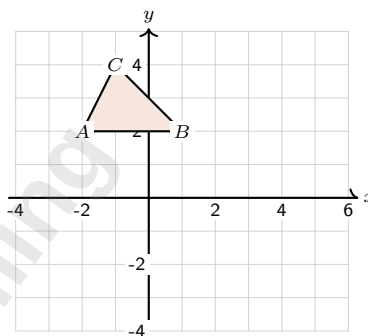
3. **Algebra Representation:** The rectangle shows  $(x + 3)(x + 2)$ . Expand the expression and label each region. [2 marks]



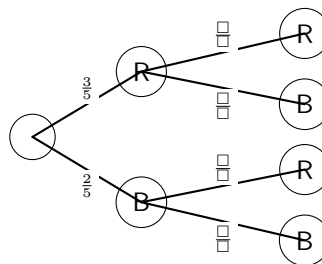
4. **Geometry:** Find  $x$  to one decimal place. [2 marks]



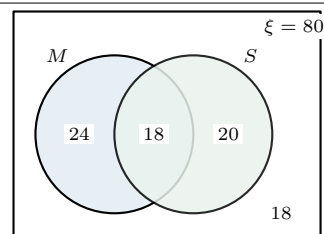
5. **Transformations:** Triangle  $ABC$  is translated by vector  $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ . Draw the image and write  $A'$ ,  $B'$ ,  $C'$ . [3 marks]



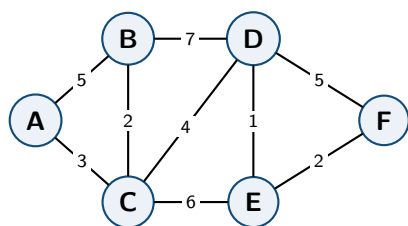
6. **Probability Tree:** A bag has 3 red and 2 blue counters. Two counters are drawn without replacement. Complete the tree and calculate  $P(\text{two red})$ . [3 marks]



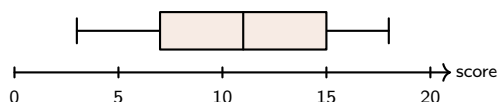
7. **Set Theory:** Use the Venn diagram to find  $n(M \cap S)$  and  $P(S | M)$ . [3 marks]



- 8. Networks:** Determine the shortest path from *A* to *F* and its total weight. **[3 marks]**



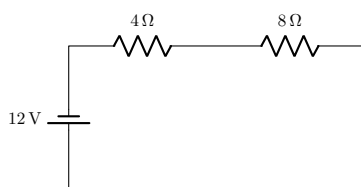
- 9. Statistics:** The box plot summarises quiz scores out of 20. State the median and interquartile range. **[2 marks]**



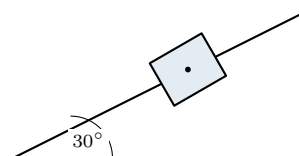
- 10. Science Investigation Skills:** A student tests whether salt affects plant growth. They use different seeds, different pots, and different window positions. Identify the independent variable, dependent variable, and two controlled variables that should be kept the same. **[4 marks]**

| Variable type         | Response |
|-----------------------|----------|
| Independent variable  |          |
| Dependent variable    |          |
| Controlled variable 1 |          |
| Controlled variable 2 |          |

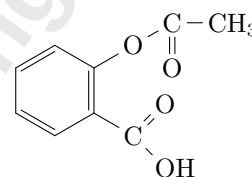
- 11. Physics - Circuits:** For the circuit below, calculate the total resistance and current if  $V = 12\text{ V}$ . **[3 marks]**



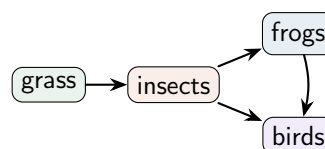
- 12. Physics - Free Body Diagram:** Draw and label the weight, normal reaction, and friction force acting on the block. **[3 marks]**



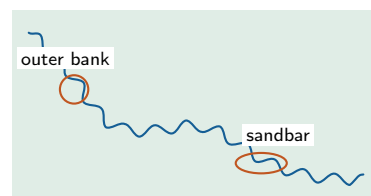
- 13. Chemistry:** Identify two functional groups in aspirin and write the equation for salicylic acid reacting with acetic anhydride to form aspirin and ethanoic acid. **[4 marks]**



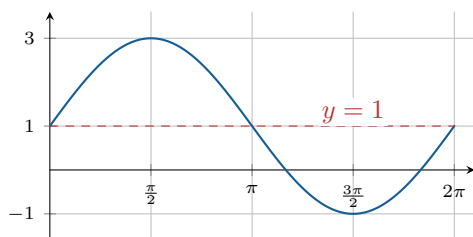
- 14. Biology - Food Web:** If the insect population suddenly falls, explain the likely short-term effect on frogs and birds. **[3 marks]**



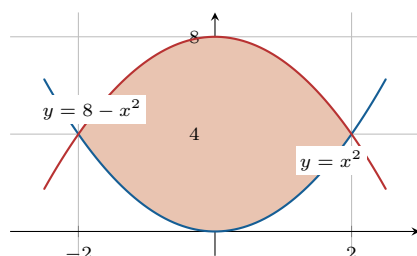
- 15. Earth Science:** Label one example each of erosion, transport, and deposition on the river diagram. **[3 marks]**



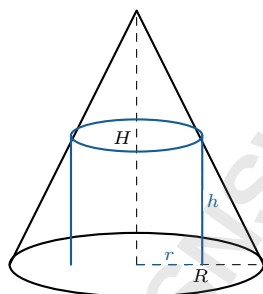
- 16. Trigonometry Graphing:** For  $y = 2\sin(x) + 1$ , state the amplitude, midline, and range. [3 marks]



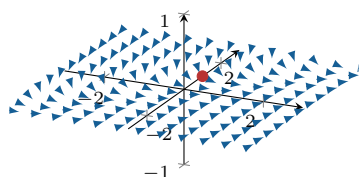
- 17. Calculus - Area:** Find the exact area bounded by  $y = x^2$  and  $y = 8 - x^2$ . [4 marks]



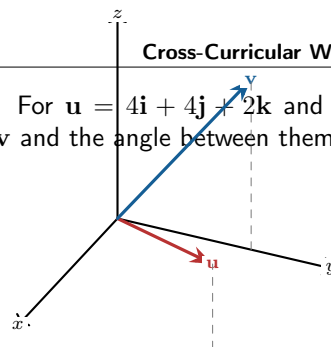
- 18. Calculus - Parameter Shift:** A cylinder is inscribed in a cone. Explain how increasing  $R$  while holding  $H$  fixed affects the maximum volume. [3 marks]



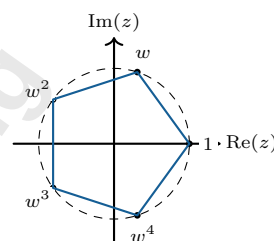
- 19. Differential Equations:** Sketch the solution curve through  $(0, 1)$  for  $\frac{dy}{dx} = x - y$ . [3 marks]



- 20. 3D Vectors:** For  $\mathbf{u} = 4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$  and  $\mathbf{v} = 3\mathbf{j} + 4\mathbf{k}$ , calculate  $\mathbf{u} \cdot \mathbf{v}$  and the angle between them. [4 marks]



- 21. Complex Numbers:** The diagram shows the fifth roots of unity. Prove  $1 + w + w^2 + w^3 + w^4 = 0$ . [3 marks]

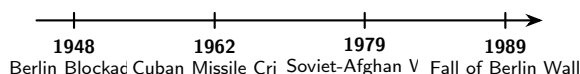


- 22. Business Studies - Critical Reasoning:** A company has already spent \$500 million on an old engine design. Electric vehicles then become cheaper to produce. The board says, "We must launch it because we already paid for it." Name the fallacy and explain the better decision rule. [3 marks]

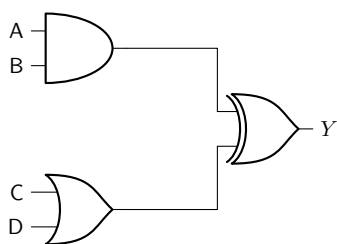
- 23. English - Literary Analysis:** Identify the main technique in the phrase "a sea of troubles" and explain its effect. [3 marks]

*Whether 'tis nobler in the mind to suffer the slings and arrows of outrageous fortune, or to take arms against a sea of troubles...*

- 24. History - Timeline Analysis:** Choose one event and explain one long-term consequence. [3 marks]



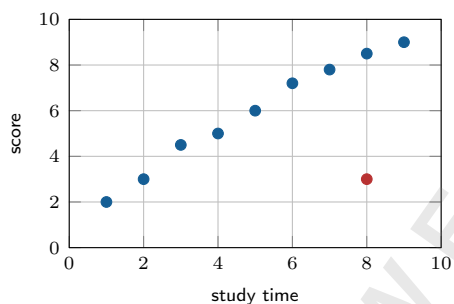
- 25. Computing - Digital Logic:** Write the Boolean expression for  $Y$ . [2 marks]



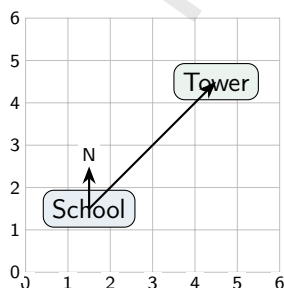
- 26. Algorithm Trace:** The recurrence is  $a_1 = 3$  and  $a_n = 2a_{n-1} - 1$ . Complete the trace table to find  $a_4$ . [2 marks]

|       |   |   |   |   |
|-------|---|---|---|---|
| $n$   | 1 | 2 | 3 | 4 |
| $a_n$ | 3 |   |   |   |

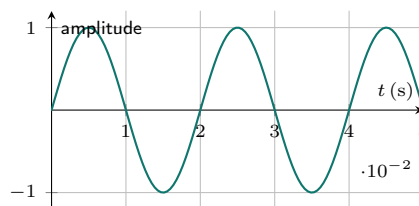
- 27. Data and Modelling:** Describe the association in the scatterplot and identify one possible outlier. [3 marks]



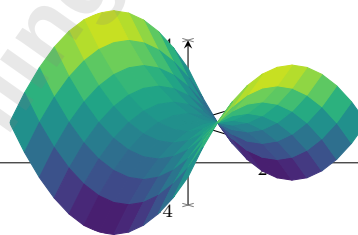
- 28. Geography - Grid Reference:** Give the grid reference for the water tower and estimate the bearing from the school to the tower. [3 marks]



- 29. Music/Physics - Waves:** The sound wave below has a period of 0.02 s. Calculate its frequency. [2 marks]



- 30. Extension - Multivariable Visualisation:** The surface is  $z = x^2 - y^2$ . Mark whether the point  $(1, 1, 0)$  lies on the surface and explain why. [2 marks]



## Part B: Year 2 Measurement Demo

Primary-level visual engagement using pure TikZ: words, objects, informal units, rulers and hands-on measuring.

### Learning Intention

I can compare and measure length using informal units and centimetres. I can explain why starting at zero matters when using a ruler.

1. **Match the words.** Draw a line from each word to the matching picture.

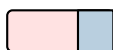
String



Rubber



Tall



2. **Word bank.** Use the words to finish the sentence.

string tall rubbers

To find out how \_\_\_\_\_ my tower is, I can measure it with \_\_\_\_\_.

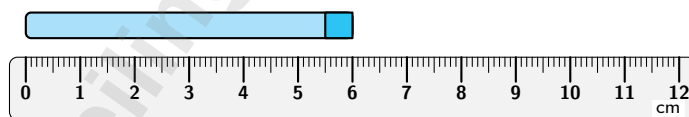
3. **How many rubbers long is the shoe?**



RUBBER RUBBER RUBBER RUBBER RUBBER

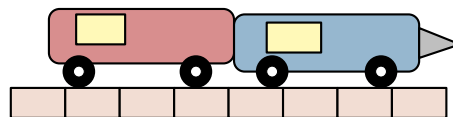
The shoe is \_\_\_\_\_ rubbers long.

4. **Use the ruler.** Measure the marker in centimetres.



The marker is \_\_\_\_\_ cm long.

5. **Toy train challenge.** Count the cubes under the train.

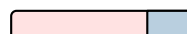


The train is \_\_\_\_\_ cubes long.

6. **Which is longer?** Circle the longer object.



pencil



rubber

## Teacher Answer Key and Marking Notes

- Q1.** 8 blocks. Award 1 for correct count from start to tip.
- Q2.** Error: the object starts at 2 cm, not zero. Correct length:  $8 - 2 = 6$  cm if reading the cap end at 8.
- Q3.**  $(x+3)(x+2) = x^2 + 5x + 6$ . Regions:  $x^2$ ,  $3x$ ,  $2x$ ,  $6$ .
- Q4.**  $x = \sqrt{6^2 + 4^2} = \sqrt{52} \approx 7.2$  m.
- Q5.**  $A'(1, 0)$ ,  $B'(4, 0)$ ,  $C'(2, 2)$ .
- Q6.** After red first:  $2/4$  red,  $2/4$  blue.  $P(RR) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$ .
- Q7.**  $n(M \cap S) = 18$ .  $P(S | M) = \frac{18}{24+18} = \frac{3}{7}$ .
- Q8.** Shortest path  $A \rightarrow C \rightarrow D \rightarrow E \rightarrow F$  with total  $3+4+1+2 = 10$ .
- Q9.** Median = 11. IQR =  $15 - 7 = 8$ .
- Q10.** IV: salt amount/concentration. DV: plant growth. Controls: seed type, pot, soil, light, water, time.
- Q11.** Series resistance  $R_T = 4 + 8 = 12 \Omega$ . Current  $I = V/R = 12/12 = 1$  A.
- Q12.** Weight vertically down, normal perpendicular to plane, friction parallel to plane opposing likely motion.
- Q13.** Functional groups: carboxylic acid and ester; aromatic ring also acceptable. Equation:  $C_7H_6O_3 + C_4H_6O_3 \rightarrow C_9H_8O_4 + C_2H_4O_2$ .
- Q14.** Frogs likely decrease because they have less food. Birds may decrease if they depend on insects/frogs, but the effect may lag or depend on alternative food.
- Q15.** Erosion: outer bank; transport: moving river load; deposition: sandbar/inside bend.
- Q16.** Amplitude 2, midline  $y = 1$ , range  $[-1, 3]$ .
- Q17.** Intersections  $x = \pm 2$ . Area =  $\int_{-2}^2 (8 - 2x^2) dx = \frac{64}{3}$  square units.
- Q18.** Since  $V_{\max} = \frac{4}{27}\pi R^2 H$ , increasing  $R$  while holding  $H$  fixed increases maximum volume quadratically.
- Q19.** Sketch should pass through  $(0, 1)$  and initially slope  $-1$ , then curve toward the solution behaviour of  $y = x - 1 + 2e^{-x}$ .
- Q20.**  $\mathbf{u} \cdot \mathbf{v} = 0 + 12 + 8 = 20$ .  $|\mathbf{u}| = 6$ ,  $|\mathbf{v}| = 5$ , so  $\cos \theta = 20/30$  and

$$\theta \approx 48^\circ.$$

$$\text{Q21. } w^5 = 1 \text{ and } w \neq 1, \text{ so } 1 + w + w^2 + w^3 + w^4 = \frac{1-w^5}{1-w} = 0.$$

**Q22.** Sunk cost fallacy. Decision should depend on future marginal costs, benefits, opportunity cost, and market fit, not unrecoverable spending.

**Q23.** Metaphor. It turns mental struggle into an overwhelming physical image, emphasising distress and scale.

**Q24.** Open response. Examples: Cuban Missile Crisis intensified nuclear caution and contributed to later arms-control communication.

$$\text{Q25. } Y = (A \wedge B) \oplus (C \vee D).$$

$$\text{Q26. } a_2 = 5, a_3 = 9, a_4 = 17.$$

**Q27.** Strong positive association; possible outlier  $(8, 3)$ .

**Q28.** Tower approximately  $(4.5, 4.5)$  or grid square around  $E5$  depending on local convention. Bearing from school to tower is about  $045^\circ$ .

$$\text{Q29. } f = 1/T = 1/0.02 = 50 \text{ Hz.}$$

$$\text{Q30. Yes. If } x = 1 \text{ and } y = 1, \text{ then } z = 1^2 - 1^2 = 0.$$

**Year 2 Q1.** Teacher checks matching: String to wavy line, Rubber to eraser, Tall to vertical height arrow.

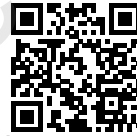
**Year 2 Q2.** "tall" and "rubbers".

**Year 2 Q3.** About 4 rubbers if aligned to the displayed informal units.

**Year 2 Q4.** 6 cm.

**Year 2 Q5.** 8 cubes.

**Year 2 Q6.** Pencil is longer.



Generate more



ISNSW example context

Name:

Date:

**My Measurement Challenge!***Learning to measure with cubes and rulers***Part 1: Match the Words**

Draw a line from the word to the matching picture.

String

Rubber

Tall



1. **Word Bank Challenge:** Use the words below to finish the sentence.

string   tall   rubbers

To find out how

my tower is, I can measure it with

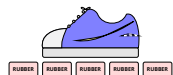
**Part 2: Hands-On Non-Standard Units***Grab your measuring tools and let's get to work!*

2. **Measure Your Desk!**

Place your hands flat on your desk, side-by-side. Measure how long your desk is from one side to the other.

My desk is                  hands long.

3. How many rubbers long is this shoe?

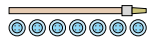


The shoe is                  rubbers long.

Page 1

**Differentiated Challenges***Extra tasks for measuring with blocks and rulers!***Level 1: Foundation (Easier)**

9. How many **buttons** long is the paintbrush?



The paintbrush is                  buttons long.

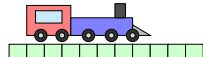
10. Measure the **Glue Stick** with the ruler below.



The glue stick is                  cm.

11. The glue stick is ( **shorter** / **longer** ) than the paintbrush.

12. How many **blocks** long is the toy train?



The toy train is                  blocks long.

13. Which word is a **unit** of length? Circle the correct answer.

**Triangle****Metre****Green**

14. Mr Darra measured this pencil with blocks. Did he do it correctly? Circle YES or NO.

**YES****NO**

Page 3



says: "3 is bigger than 2, so my Glue Stick must be longer!"

**Explain why we cannot measure one thing with rubbers and another thing with pencils to compare their sizes:**

**Reading for Fluency***Let's practice reading like experts!***The 3 Keys to Good Reading**

- 1. Accuracy:** We sound out tricky words instead of guessing.
- 2. Expression:** We don't read like a robot! Our voice goes up at a question mark (?) and we take a breath at a full stop (.).
- 3. Understanding:** We make a movie in our head about what the words mean.

22. **Paired Reading:** Read this story to your partner using the 3 Keys.

Today, Mr Darra showed us how to measure. We learned that we must use the same unit to compare sizes! If uses pencils and uses rubbers, we get confused. raised his hand and asked, "But what if my pencils are short?" Mr Darra smiled. "That is a great question!" he said. Then, asked, "Can we just use a ruler instead?" We all agreed that rulers make measuring fair. Do you know how to use a ruler?

23. **My Reading Checklist:** Tick the boxes after you read!

- ☐ I sounded out the tricky words.
- ☐ I stopped at the full stops.
- ☐ I know what the sentence means.

Page 5

**Part 3: Using a Ruler (cm)**

4. Use the ruler below to measure the marker in centimetres (cm).



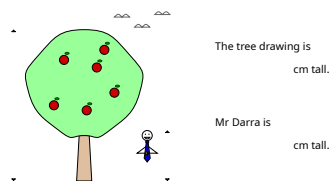
The marker is                  cm long.

5. **Level 2:** This battery does **not** start at 0. Find the length.



Starts at:                  Ends at:                  Length:                  cm.

6. **Tree Challenge:** Measure the **height** of the tree and Mr Darra.



The tree drawing is                  cm tall.

Mr Darra is                  cm tall.

**Part 4: Length Logic**

7. **Ordering (Baseline):** Order these lengths from **shortest** to **longest**.

18 cm,    64 cm,    42 cm,    110 cm

Shortest:                  →                  →                  → Longest:

8. **Estimation:** How tall is Ms                  Circle one.

170 cm    17 cm    17 m

Page 2

**Level 2: Intermediate (Medium)**

15. Find the total length if you put the marker and the battery together.

*Hint: Marker = 6 cm, Battery = 3 cm. What is the total?*

Working:                  Total:                  cm.

16. Write a word from your word bank that can be used to measure a curved line.

Answer:

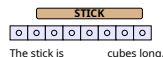
**Level 3: Extension & Reflection**

17. **Super Challenge (Mixed Units):** Order these lengths from shortest to longest.

15 cm,    3 m,    110 cm,    1 m,    80 cm

Shortest:                  →                  →                  → Longest:

18. This stick does **not** start at the first cube. How many cubes long is it?



The stick is                  cubes long.

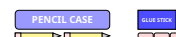
19. Write a sentence explaining **how** you found the length of the stick when it didn't start at the beginning.

20. **Word Problem Challenge:** A piece of wood is 1 metre long. Mr Darra cuts off 20 cm. How long is the wood now?

Working:                  Answer:                  cm

21. **The Measurement Trap!** Mr Darra asked two students to measure items.

- measured a Pencil Case. It was **2 pencils** long.
- measured a Glue Stick. It was **3 rubbers** long.



Page 4



24. **Writing: Teach an Alien!**

Write 3 sentences to teach an alien how to measure your desk.

First, I

Next, I

Finally, I

25. **Reflection:** What was the trickiest part about measuring today?

I thought                  was tricky because

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